

Numerical Method (CSC212)

B.Sc. CSIT · Third Semester · Tribhuvan University
Most Important Questions & Topics (Based on 2075–2081 Past Papers)

■■■■ Very High Priority

■■■ High Priority

■ Moderate Priority

UNIT 1: Solution of Nonlinear Equations

1.1 Newton-Raphson Method ■■■■

■ Appeared in: 2081, 2077

Q1. Derive the formula for Newton-Raphson method. Solve $x^2+4x-9=0$ with error precision 0.01. Discuss its drawbacks.

Q2. What are inherent errors? Derive Newton-Raphson method and solve $x^2-5x+6=0$ up to 3 decimal places.

Q3. Why does Newton-Raphson method fail sometimes? Explain with example.

1.2 Secant Method ■■■■

■ Appeared in: 2080, 2079, 2075

Q1. How does Secant method differ from Newton-Raphson? Derive the formula and solve $\cos(x)+2\sin(x)-x^2=0$ with error 0.01.

Q2. Derive Secant method formula and find root of $x^3+2x-\cos(x)=4$ with error precision 0.01.

Q3. Find root of $x^2+\cos(x)-e^{(-x)}-2=0$ using Secant method.

1.3 Half-Interval (Bisection) Method ■■

■ Appeared in: 2078, 2080, 2079

Q1. How can bisection method estimate root of a non-linear equation? Find root of $x^2-e^{(-x)}-x=1$ up to 2 decimal places.

Q2. Find real root of $x^2-e^{(-x)}=3$ using bisection method up to 3 significant figures.

Q3. Define approximate error and relative approximate error. Discuss bisection method.

1.4 Fixed Point Iteration ■■

■ Appeared in: 2078, 2079

Q1. Calculate real root of $2x^3-2x=5$ using fixed point iteration up to 3 significant figures.

Q2. What is fixed point iteration? Explain convergence and divergence cases with examples.

1.5 Horner's Method ■

■ Appeared in: 2078, 2077

Q1. How can Horner's rule evaluate $f(x)$ and $f'(x)$ of a polynomial? Write algorithm and program for real root using Horner's method.

Q2. Use Horner's method to evaluate $2x^3-3x^2+5x-2$ at $x=3$. Write its algorithm.

UNIT 2: Interpolation and Regression

2.1 Newton's Interpolation (Divided/Forward/Backward Differences) ■■■

■ Appeared in: 2081, 2080, 2078, 2077

Q1. What is Newton's interpolation? Build divided difference table and estimate $f(2)$ and $f(5)$ from given data.

Q2. Estimate $f(0.6)$ using Newton's interpolation formula from given data points.

Q3. Define interpolation and extrapolation. Write algorithm and program for Newton's divided difference interpolation.

Q4. Construct Newton's backward difference table and approximate $f(45)$ from given data.

Q5. Construct Newton's forward difference table and approximate $f(15)$ from given data.

2.2 Lagrange Interpolation ■■■

■ Appeared in: 2081, 2077

Q1. Write algorithm and program to implement Lagrange interpolation method.

Q2. How does interpolation differ from regression? Write algorithm and program for Lagrange interpolation.

2.3 Cubic Spline Interpolation ■■

■ Appeared in: 2081, 2080, 2079

Q1. Given data points X : 1.0, 3.0, 4.0 and $f(x)$: 1.5, 4.5, 9.0 — find cubic spline for $1 \leq x \leq 3$ and estimate $f(2)$.

Q2. How does spline interpolation differ from Lagrange's? Estimate $f(0)$ and $f(4)$ using cubic spline from given data.

Q3. What are problems with polynomial interpolation for large data sets? How are they addressed?

2.4 Linear and Non-linear Regression ■■

■ Appeared in: 2081, 2078, 2079, 2077, 2075

Q1. What is linear regression? Fit a linear function to given data points.

Q2. What is regression analysis? Fit a second order polynomial for given data values.

Q3. Fit quadratic curve through given data points and estimate $f(2)$ using least squares.

Q4. Fit the curve $y = ae^{bx}$ through given data points.

Q5. Fit a function of type $y = a + bx$ using least squares method.

UNIT 3: Numerical Differentiation and Integration

3.1 Numerical Integration — Simpson's Rules ■■■

■ Appeared in: 2081, 2079, 2078, 2075

Q1. Find value of $\int (e^x/x) dx$ from 1 to 2 using Simpson's 3/8 rule with $n=6$.

Q2. Estimate integral value of given tabulated function using Simpson's 1/3 rule ($x=1.2$ to 2.4).

Q3. How does Simpson's 1/3 rule differ from Trapezoidal rule? Derive Simpson's 1/3 formula.

Q4. Find integral value from $x=1.8$ to 3.4 using Simpson's 1/3 rule.

3.2 Romberg Integration / Gaussian Integration ■■

■ Appeared in: 2078, 2079, 2075

- Q1. Evaluate $\int (\sin^2 x / x) dx$ from 0 to 1 using Romberg integration.
- Q2. Evaluate $\int (\sin(x)/x) dx$ from 0 to 1 using Romberg integration.
- Q3. Evaluate $\int (\sin(x)/x) dx$ from 0 to 1 using Gaussian integration three-ordinate formula.

3.3 Trapezoidal Rule ■■

■ Appeared in: 2080

- Q1. Find $\int (x^3+2) dx$ from 2 to 4 using composite trapezoidal rule for $k=2$ and $k=4$ segments.

3.4 Numerical Differentiation ■

■ Appeared in: 2081, 2077

- Q1. What is numerical differentiation? Estimate velocity and acceleration at $x=4$ from tabulated distance-time data.
- Q2. Write algorithm and program for differentiating continuous functions using three-point formula.

UNIT 4: Solving System of Linear Equations

4.1 Gauss-Jordan / Gauss Elimination Method ■■■

■ Appeared in: 2081, 2080, 2078, 2077

- Q1. Solve given system of 4 equations using Gauss-Jordan elimination method.
- Q2. How does Gauss-Jordan differ from Gauss Elimination? Solve 3×3 system using Gauss-Jordan. Also find inverse of matrix.
- Q3. Why is partial pivoting used with naive Gauss Elimination? Solve system using Gauss Elimination with partial pivoting.
- Q4. Write algorithm and program to solve n linear equations using Gaussian elimination.

4.2 LU / Matrix Factorization (Doolittle & Cholesky) ■■■

■ Appeared in: 2078, 2075, 2077

- Q1. What is matrix factorization? Factorize matrix A using LU (Doolittle) and solve $Ax=b$.
- Q2. Factorize the given 4×4 matrix A into LU and solve $Ax=b$ using L and U .
- Q3. Discuss the Doolittle LU decomposition method for matrix factorization.
- Q4. Factorise the given matrix using Cholesky method.

4.3 Iterative Methods: Gauss-Seidel & Jacobi ■■■

■ Appeared in: 2081, 2079, 2078, 2075

- Q1. What are limitations of direct methods? How does Gauss-Seidel differ from Jacobi? Solve given system using Jacobi iteration.
- Q2. Solve system of equations using Gauss-Seidel method (various problems from past papers).
- Q3. Solve the given 3-equation system using Gauss-Seidel method.

UNIT 5: Solution of Ordinary Differential Equations

5.1 Runge-Kutta 4th Order Method ■■■

■ Appeared in: 2081, 2079, 2075

- Q1. Solve $dy/dx=xy$, $y(0)=1$ at $x=0.4$ using RK 4th order method.
- Q2. Solve $y'(x)+y=3x$ with $y(0)=2$ for $0 \leq x \leq 1$, $h=0.5$ using RK 4th order method.
- Q3. Estimate $y(1)$ from $dy/dx+2x^2y=4$, $y(0)=1$ using RK 4th order ($h=0.5$).

5.2 Heun's Method ■■■

■ Appeared in: 2080, 2079, 2078

- Q1. Solve $y'(x)+x^2y=3x$ with $y(1)=1$ for $1 \leq x \leq 2$, $h=0.25$ using Heun's method.
- Q2. Write program for solving ODE using Heun's method.

5.3 Euler's Method & Taylor Series ■■

■ Appeared in: 2077, 2080

- Q1. Approximate $y'=2x+y$, $y(0)=1$ using Euler's method ($h=0.1$). Approximate $y(0.4)$.
- Q2. Approximate $y'=3x^2$, $y(1)=1$ using Taylor's series with first four terms. Find $y(2)$.

5.4 Higher-Order ODEs & Boundary Value Problems ■■

■ Appeared in: 2078, 2075, 2077

- Q1. Solve $d^2y/dx^2+3(dy/dx)+5y=0$ for $1 \leq x \leq 2$, $h=0.25$ with $y(1)=1$, $y'(1)=2$.
- Q2. What is a boundary value problem? Write algorithm and program to solve BVP using shooting method.
- Q3. How do BVPs differ from IVPs? Discuss shooting method.

UNIT 6: Solution of Partial Differential Equations

6.1 Poisson's / Laplace Equation (Plate Temperature Problems) ■■■

■ Appeared in: 2081, 2080, 2079, 2078, 2077, 2075

- Q1. Solve Poisson equation $\partial^2u/\partial x^2+\partial^2u/\partial y^2=-64xy$ on $[0,1] \times [0,1]$ with given boundary conditions ($h=1/3$).
- Q2. Solve $\nabla^2f=xy$ with $f=2$ on boundary, square domain $0 \leq x,y \leq 3$, $h=1$.
- Q3. Solve $\nabla^2f=3x^2y$ over $0 \leq x,y \leq 3$, $f=0$ on boundary, $h=1$.
- Q4. Solve Poisson's equation over $0 \leq x,y \leq 1.5$, $f=0$ on boundary, $h=0.5$.
- Q5. 90cm×90cm metallic plate: two sides at 100°C, two at 200°C, grid 30cm — find interior temperatures.
- Q6. 18cm×18cm plate: left=100°C, right=200°C, upper=50°C, lower=150°C, grid 6cm — find interior node temperatures.

■ Exam Strategy & Tips

Group A (Long Questions)	Expect 1 from Unit 1 (Newton-Raphson/Secant), 1 from Unit 2 (Interpolation/Algorithm), and 1 from Unit 4 (Linear Equations). Answer any 2.
Group B (Short Questions)	Highly repeated: Gauss-Seidel/Jacobi, RK-4, Heun's method, Poisson's equation plate temperature, Simpson's rule, Lagrange/Newton interpolation.

Unit 6 (PDE)	Plate temperature problems appear in EVERY year's paper. Practice at least 3 variants with different boundary conditions.
Algorithm/Program Questions	Always prepare algorithms for: Newton-Raphson, Lagrange, Gauss Elimination, Heun's method, RK-4. These are frequently asked.
Most Repeated Topics	1) Gauss-Seidel/Jacobi 2) RK-4 3) Poisson plate problems 4) Newton/Secant method 5) LU Factorization

Based on TU past papers 2075–2081 | CSC212 Numerical Method | B.Sc. CSIT Semester III